

Woodgas Donor Vehicle

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Choosing a woodgas donor vehicle is one of the most important decisions you'll make. There are several factors involved. This has been covered over and over on the forum, so I've gathered up most of the answers here for easy reference.

Note: Using what you've already got is perfectly acceptable. However please compare it to this list and adjust your expectations accordingly.

Highway Burner

This would be your distance traveling vehicle, able to keep up with freeway traffic, comfortable for long trips. Good handling, good stopping, and general road-worthiness are priorities.

While passenger cars would be more comfortable than a truck, we generally use pickups to make installing easier. You can apply these criteria to other vehicles if you have alternative mounting in mind.

Old school V8 engine

We like big pushrod V8 engines for simplicity and reliability. They make power at low RPMs, have plenty of torque, offer performance upgrades, and generally provide more working space under the hood. The V8 configuration provides 45 degrees timing advance above stock, which is exactly right.

There are some modern pushrod V8s which may be worth a look. The LSx engines in particular follow a long tradition of compact powerful engines, while making more power out of less fuel than the older generations did. Watch this space...

Light weight & low profile

The less weight, the better performance. A full size truck is a heavy burden to move, due to it's large profile and weight. Small trucks slip through the wind and weigh significantly less. However it's hard to find a pushrod V8 in a small truck. Some conversions have been done, like V8 S10s.

Watch closely on these light trucks for rust. With all that weight savings, you can't afford to lose the steel that's left. If the frame is questionable, you need to keep looking.

Stays on the road

Trucks optimized for highway travel won't have 4WD, high clearance, large tires, etc. These work against efficiency and speed. If you get one that's also ready for offroad, you'll be facing compromises on the highway. We really want a truck that's a car underneath.

Australians have formalized this car/truck idea into an actual vehicle called a Ute. Front half car, rear half truck - perfect for woodgas! Unfortunately we can't buy them in the States.

EXAMPLES: Dodge Dakota, S10 V8 conversion, Ranchero / El Camino, Ute

Work Truck

This is the classic farm truck that needs to move heavy loads at low speeds, but rarely used for trips, and isn't expected to keep up with freeway traffic. There's a lot more room in this category, since speed is out of the equation. Mostly "bigger is better". Heavy duty rules the day, but you can use what you bring. I'm guessing that a lot of you already have a truck in this category.

Biggest engine you can get

Opt for a V10 if you can get it, or a very large V8. In this case displacement trumps all else. The Dodge Ram V10 has an 8 liter engine, the largest available in a pickup truck.

Large engines we know work well:

- Dodge V10 8L (488 ci)
- Ford V8 7.5L (460 ci)
- Chevy V8 7.4L (454 ci)

Heavy & high profile is OK

You'll probably end up with a 6-7,000 pound truck, that stands tall in the wind. Since you're moving slow and have a monster engine, this doesn't matter as much. You'll want a taller gasifier with more hopper capacity on this beast. Making the cooler tall as well is not a problem. It all goes hand in hand with the larger loads and engine size.

Go slow Joe

I'll emphasize again, this won't be a fast truck! Expect to maintain 55-60 unloaded. Less with a full load. If you need to go faster than that, you can try running hybrid with some gasoline. It's not driving for free, but it will still save you a bundle. Our tests seem to show the minimum you can run is about 10-20% gasoline. That's a significant speed boost.

EXAMPLES: Dodge Ram V10, Ford F250 with 460 V8, Chevy 2500 454

All trucks need:

Pickup bed

Defined as an open air bed directly behind the driver, with a wall in between. This allows the best mounting position and storage options. Variations include El Camino/Racheros, utes, custom built flatbeds, modified vans/SUVs. Don Mannes actually converted a Geo Tracker into a pickup, with great results!

Overdrive

Keep those RPMs down! An extra gear means another 5-10 MPH on the highway, or better efficiency at the same speed. This is a mark against older trucks, which often have only 3 speeds and no overdrive gear. Transmission swaps are fairly simple for some models, if you want to upgrade your ride.

Gasoline

Use gasoline engines, not diesels. Diesels are harder to convert and the challenges usually outweigh the advantages. Again I will emphasize, we do NOT currently support using diesels.

That said, diesel conversions are possible, and quite intriguing. Eventually I hope an experienced woodgasser tries converting one. But only as your third or fourth truck, with many thousands of woodgas miles under your belt.

Fuel injection

The easiest way to switch from gasoline to woodgas is by shutting off an injector. Carburetors do not shut off easily, nor do they stop "dribbling" once woodgas is in use. They must be drained each time to completely stop the gasoline. Plus you are faced with either going through the carb with the woodgas and plugging small passageways, or going under and dealing with leaks and dual throttles.

Carburetors are not a deal-breaker. I've run them with no issues. In some situations they even have advantages. But in general we advise against them.

MPFI

Of the methods used in fuel injection, Multi-port fuel injection (MPFI) is much preferable to Throttle-body injection (TBI). The reasons are many; MPFI engines have more HP than TBI, MPFI intakes have smooth clean-air passageways, TBI intakes create turbulent flow to entrain the gasoline droplets. MPFI allows easy cleaning of the carbon buildup in the intake; TBI must be disassembled and cleaned by hand.

OBD1

Vehicle computers have gotten smarter lately, and this can make woodgas conversions difficult. The original OBD1 systems are simple and work very well with woodgas. OBD2 might work, and it might cause problems, depending on the model.

Some early models of OBD2 are very similar to the last of the OBD1 systems. In particular, the Dakotas 1996-1998 behave practically the same as the earlier trucks. This makes them good candidates.

Space

Look for space to run pipes. The Dakota has 11" under the bed, whereas most trucks have 4-5". Older square-body trucks are simpler under the hood and have more room. High power-weight ratio sometimes means a big engine under a small hood, so that can cramp your plumbing. This alone can rule out some V8 swaps into smaller vehicles - by the time you're done, you have no room for the extra woodgas equipment.

Capacity

Make sure that the vehicle will be useful to you. Trucks often only seat a couple passengers. Cargo area will be slightly diminished. Towing ability will be limited. Can you still do what you need to do?

This is especially true of trailer mounted gasifiers. It means you can't tow anything else. Bumper mounting probably means no trailer unless you're quite clever.

About Drive On Wood

A community of gasification enthusiasts hosted by Wayne Keith, resident expert & woodgas pioneer.

Drive On Wood is a partnership of Wayne Keith & Chris Saenz.

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